



GOSFORD HIGH SCHOOL

2010/2011 HIGHER SCHOOL CERTIFICATE

MATHEMATICS EXTENSION 2

ASSESSMENT TASK 1

Time Allowed – 60 minutes

- All necessary working should be shown.
- Full marks may not be awarded for unnecessarily untidy work or work that is poorly organised.
- Students must begin each new question on a new sheet of paper.
- Questions will be collected separately at the conclusion of the assessment task.
- All questions are to be attempted and are of equal value.

Question 1 - (15 marks) – Use a separate sheet of paper

Marks

a. Let $z = 3 + 4i$ and $w = -1 + i$

Express the following in the form $a + ib$, where a, b are real numbers:

1

i. $z - w$

1

ii. iz

1

iii. $z + \bar{z}$

2

iv. $\frac{z}{w}$

2

v. w^4

b. If $\alpha = \frac{5 (\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})}{2 (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})}$

1

Express α in mod - arg form.

c. Let $\beta = -2 + 2i\sqrt{3}$

i. Find the exact value of $|\beta|$ and $\arg\beta$.

2

ii. Hence, express β^5 in the form $a + ib$, where a, b are real numbers.

2

d. Find the complex square roots of $7 + 6\sqrt{2}i$.

3

End of Question 1

Question 2 - (15 marks) – Use a separate sheet of paper

Marks

- a. 1
- i. On an Argand diagram show vectors representing z_1 , z_2 , $z_1 + z_2$ and $z_1 - z_2$. 2
- ii. If z_1 and z_2 are two complex numbers such that $\frac{z_1 + z_2}{z_1 - z_2} = 2i$ show that $|z_1| = |z_2|$. 2
- b. On an Argand diagram, shade the region defined by: 2
- $\text{Im}(z) \leq 2$ AND $\frac{\pi}{3} \leq \arg(z + i) \leq \frac{\pi}{2}$
- c. 2
- i. Sketch the graph on the Argand diagram specified by: 1
- $|z - 1 - \sqrt{3}i| = 2$
- ii. Hence, find the maximum value of $|z|$. 2
- d. Find the three cube roots of $8i$ (express answers in the form $a + ib$). 2
- e. 2
- i. Find all solutions of the equation $z^5 = 1$, giving your answers in modulus – argument form. 1
- ii. Show on a diagram that the points representing the five roots of this equation, on an Argand diagram, form the vertices of a regular pentagon. 2
- iii. Hence, show that the perimeter of this regular pentagon is $10 \sin \frac{\pi}{5}$ units.

End of Question 2

Question 3 - (15 marks) – Use a separate sheet of paper

Marks

- a. Sketch the region in the complex plane where the inequalities $|z| > 2$ and $2 \leq z + \bar{z} \leq 6$ hold simultaneously. 3
- b. Given $z = 3 + 4i$, find all possible co-ordinates of w , so that the origin 0 , z and w form a right angled isosceles triangle (right angled at z) on the Argand diagram. 3
- c. The point P on the Argand diagram represents the complex number $|z - 1| = \operatorname{Re}(z)$
- i. Find the equation of the locus of P in terms of x and y . 3
- ii. Describe the locus geometrically, giving at least two features of the curve.
- d. Find the locus in cartesian form of z if:
 $\arg(z + 4) = \arg(z - 4) - \frac{\pi}{4}$. 3
- e. Find the Cartesian equation of the locus of w if $w = \frac{z-2}{z}$ and given that $|z| = 2$. Describe the locus geometrically. 3

End of Assessment Task

1a. (i) $z - w = 3 + 4i - (-1 + i)$
 $= 4 + 3i$

(ii) $iz = i(3 + 4i)$
 $= -4 + 3i$

(iii) $z + \bar{z} = 3 + 4i + 3 - 4i$
 $= 6$

(iv) $\frac{z}{w} = \frac{3+4i}{-1+i} \times \frac{-1-i}{-1-i}$
 $= \frac{-3-3i-4i+4}{1+1}$
 $= \frac{1-7i}{2}$
 $= \frac{1}{2} - \frac{7}{2}i$

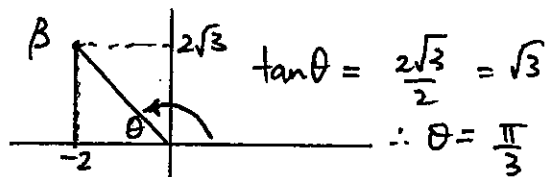
(v) $w^4 = (w^2)^2$
 $= [(-1+i)^2]^2$
 $= (1-2i-1)^2$
 $= (-2i)^2$
 $= 4i^2$
 $= -4$

b. $\alpha = \frac{5}{2} \operatorname{cis} \left(\frac{\pi}{3} - \frac{2\pi}{3} \right)$
 $= \frac{5}{2} \operatorname{cis} \left(-\frac{\pi}{3} \right)$

c. (i) $\beta = -2 + 2\sqrt{3}i$

$|\beta| = \sqrt{(-2)^2 + (2\sqrt{3})^2}$
 $= \sqrt{4 + 12}$

$= \sqrt{16}$
 $= 4$



$\therefore \arg \beta = \frac{2\pi}{3}$

(ii) $\beta = 4 \operatorname{cis} \left(\frac{2\pi}{3} \right)$
 $\beta^5 = \left[4 \operatorname{cis} \left(\frac{2\pi}{3} \right) \right]^5$
 $= 1024 \operatorname{cis} \left(\frac{10\pi}{3} \right)$
 $= 1024 \operatorname{cis} \left(-\frac{2\pi}{3} \right)$
 $= -512 - 512\sqrt{3}i$

d. Let $z = \sqrt{7 + 6\sqrt{2}i}$

$z^2 = 7 + 6\sqrt{2}i$

$\therefore (x+iy)^2 = 7 + 6\sqrt{2}i$

$x^2 - y^2 + 2xyi = 7 + 6\sqrt{2}i$

Equating real, imaginary parts

$x^2 - y^2 = 7 \quad (1)$

$2xy = 6\sqrt{2} \quad (2)$

From (2) $y = \frac{3\sqrt{2}}{x}$

In (1) $x^2 - \left(\frac{3\sqrt{2}}{x} \right)^2 = 7$

$x^2 - \frac{18}{x^2} = 7$

$x^4 - 7x^2 - 18 = 0$

$(x^2 + 2)(x^2 - 9) = 0$

Since x is real: $x = \pm 3$, $y = \pm\sqrt{2}$
the square roots of $7 + 6\sqrt{2}i$ are
 $3 + \sqrt{2}i$, $-3 - \sqrt{2}i$

d) By inspection :

one root is $-2i$ (since $(-2i)^3 = 8i$)

$-2i = 2 \operatorname{cis}(-\frac{\pi}{2})$ and the two other roots are evenly spaced around Argand diagram

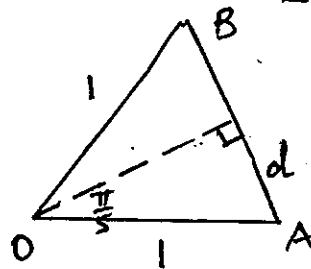
i.e $z_1 = -2i$

$$z_2 = 2 \operatorname{cis}(\frac{\pi}{6}) = 2(\frac{\sqrt{3}}{2} + \frac{i}{2}) = \sqrt{3} + i$$

$$z_3 = 2 \operatorname{cis}(\frac{5\pi}{6}) = 2(-\frac{\sqrt{3}}{2} + \frac{i}{2}) = -\sqrt{3} + i$$

$\therefore$ 3 roots are : $-2i, \sqrt{3} + i, -\sqrt{3} + i$

(iii) Consider $\triangle AOB$:



$$\frac{d}{1} = \sin \frac{\pi}{5}$$

$$\therefore d = \sin \frac{\pi}{5}$$

$$\therefore AB = 2 \sin \frac{\pi}{5}$$

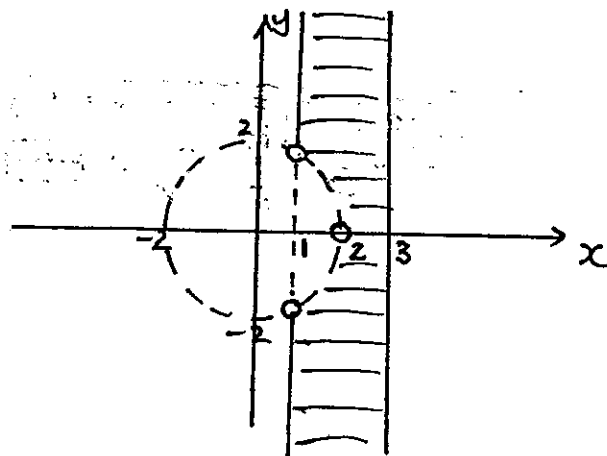
$$\therefore \text{Perimeter of pentagon} = 5 \times AB = 10 \sin \frac{\pi}{5} \text{ units as req.}$$

3)a) $|z| > 2 \rightarrow$ outside circle centre $(0,0)$ radius 2 units

$$2 \leq z + \bar{z} \leq 6 \rightarrow z + \bar{z} = x + iy + x - iy = 2x$$

i.e $2 \leq 2x \leq 6$

$$1 \leq x \leq 3$$



2)(i) If $z^5 = 1$, by inspection

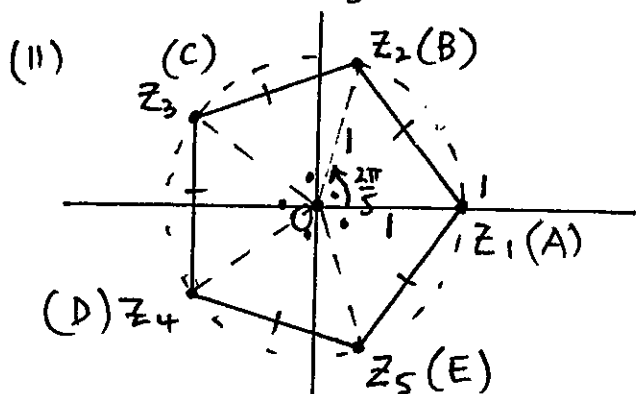
$z_1 = 1$ and other 4 are evenly spaced around Argand Diagram.

i.e $z_2 = \operatorname{cis} \frac{2\pi}{5}$

$$z_3 = \operatorname{cis} \frac{4\pi}{5}$$

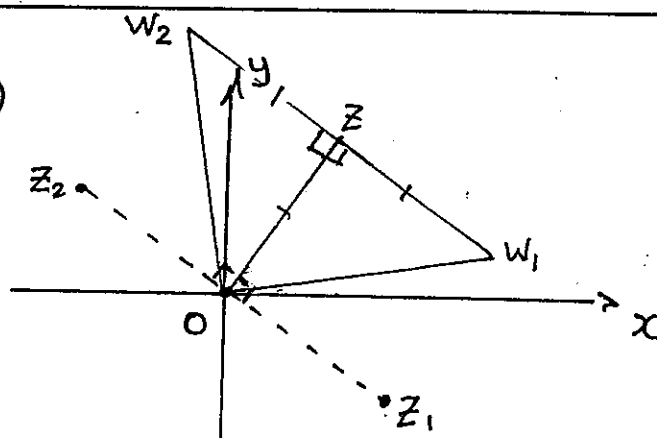
$$z_4 = \operatorname{cis}(-\frac{4\pi}{5})$$

$$z_5 = \operatorname{cis}(-\frac{2\pi}{5})$$



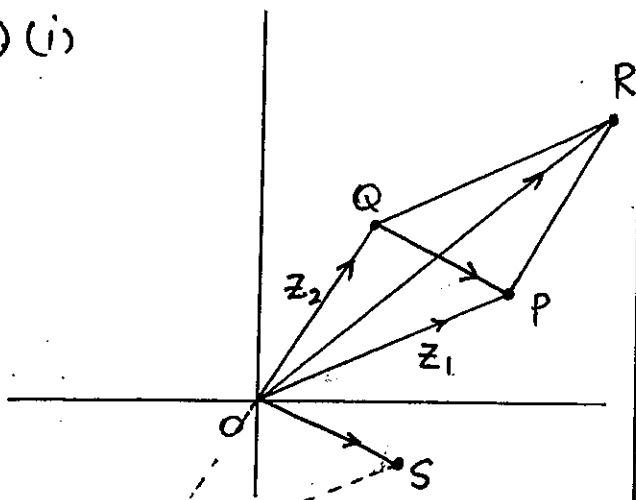
$\therefore$ A regular pentagon is formed

b)



Q2

a) (i)



Let $z_1 = \vec{OP}$

$z_2 = \vec{OQ}$

$\therefore z_1 + z_2 = \vec{OR}$

$z_1 - z_2 = \vec{OS} = \vec{QP}$

(ii) $\frac{z_1 + z_2}{z_1 - z_2} = 2i$

$\therefore z_1 + z_2 = 2i \times (z_1 - z_2)$

So $\vec{OR}$ is obtained by rotating $\vec{QP}$ 90° in an anticlockwise direction including an enlargement by a factor of 2.

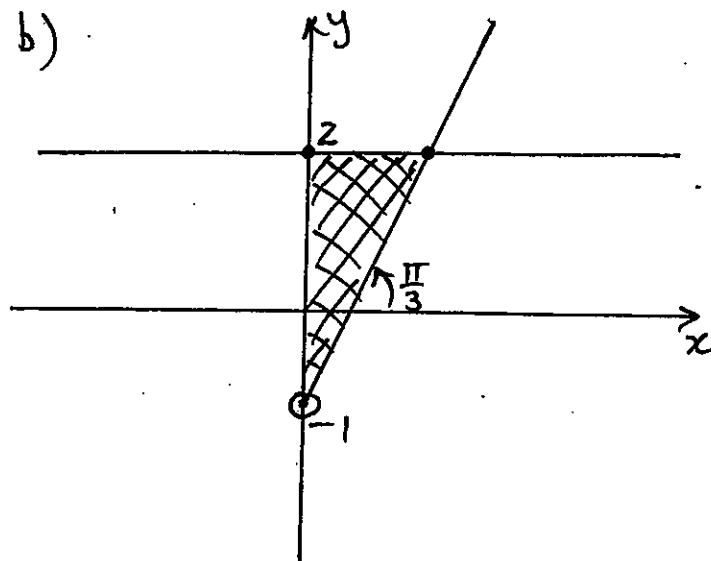
$\therefore$ diagonals of parallelogram OPRQ are at right angles to each other

$\therefore$ OPRQ is a rhombus

$\therefore \vec{OP} = \vec{OQ}$

$\therefore |z_1| = |z_2|$ as required

b)

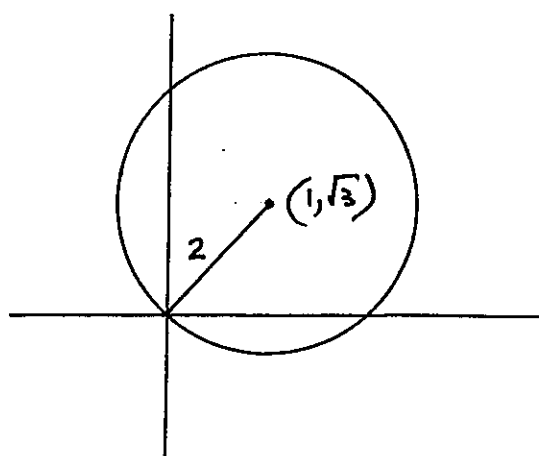


c) (i) $|z - (1 + \sqrt{3}i)| = 2$

Circle radius 2 units, centre $(1, \sqrt{3})$ and

$|1 + \sqrt{3}i| = \sqrt{1^2 + (\sqrt{3})^2} = 2$

$\therefore$ Circle passes through $(0, 0)$



(ii) Max value of $|z|$ will be the diameter of circle i.e. $2 \times 2 = 4$ units.

There are two possible positions for $w \rightarrow w_1, w_2$

$$w_1 = z + z_1$$

$$\begin{aligned} \text{where } z_1 &= z \times -i \\ &= (3+4i) \times -i \\ &= -3i + 4 \\ &= 4 - 3i \end{aligned}$$

$$\begin{aligned} \therefore w_1 &= 3+4i + 4-3i \\ &= 7+i \end{aligned}$$

$$w_2 = z + z_2$$

$$\begin{aligned} \text{where } z_2 &= z \times i \\ &= (3+4i) \times i \\ &= 3i - 4 \\ &= -4 + 3i \end{aligned}$$

$$\begin{aligned} \therefore w_2 &= 3+4i - 4 + 3i \\ &= -1 + 7i \end{aligned}$$

$$\therefore w = 7+i \text{ or } -1+7i$$

$$\left. \begin{aligned} \text{with vertex} &= \left(\frac{1}{2}, 0\right) \\ \text{focus} &= (1, 0) \\ \text{Directrix} &\Rightarrow x = \frac{1}{2} \\ \text{Axis} &: y = 0 \end{aligned} \right\} \begin{array}{l} \text{only} \\ 2 \\ \text{required} \end{array}$$

$$\begin{aligned} d) \arg(z+4) &= \arg(z-4) - \frac{\pi}{4} \\ \frac{\pi}{4} &= \arg(z-4) - \arg(z+4) \end{aligned}$$

$$\therefore \arg\left(\frac{z-4}{z+4}\right) = \frac{\pi}{4}$$

and use algebraic approach

$$\text{starting } \operatorname{Re}(z+4) = \operatorname{Im}(z-4)$$

or better to use geometric approach

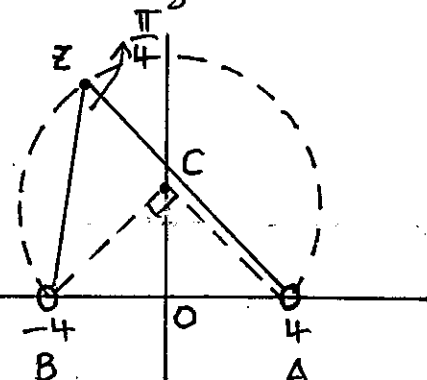
$$\arg(z-4) - \arg(z+4) = \frac{\pi}{4}$$

implies locus is major arc of circle $\rightarrow$

Since $\triangle OCB$
is right angled
isosceles
 $OC = 4$

$$\therefore C = (0, 4)$$

$$\therefore CB = \sqrt{32} = 4\sqrt{2}$$



$\therefore$ Locus is major arc of circle

$$x^2 + (y-4)^2 = 32 \text{ excluding } (-4, 0)(4, 0)$$

$$c) (i) |z-1| = \operatorname{Re}(z)$$

$$\text{let } z = x+iy$$

$$\therefore \sqrt{(x-1)^2 + y^2} = x$$

$$(x-1)^2 + y^2 = x^2$$

$$x^2 - 2x + 1 + y^2 = x^2$$

$$y^2 = 2x - 1$$

(ii) This is the equation of a parabola.

$$y^2 = 4 \cdot \frac{1}{2} \left(x - \frac{1}{2}\right)$$

$$e) w = \frac{z-2}{z} \quad \therefore 2 = \left| \frac{z}{1-w} \right|$$

$$zw = z - 2$$

(from data)

$$zw - z = -2$$

$$2 = \frac{2}{|1-w|}$$

$$z(w-1) = -2$$

$$|1-w| = 1$$

$$z = \frac{-2}{w-1}$$

$$\therefore |w-1| = 1$$

$$z = \frac{2}{1-w}$$

and locus of w is circle centre $(1, 0)$ radius 1 unit

$$\therefore |z| = \left| \frac{2}{1-w} \right|$$

$$\therefore (x-1)^2 + y^2 = 1$$